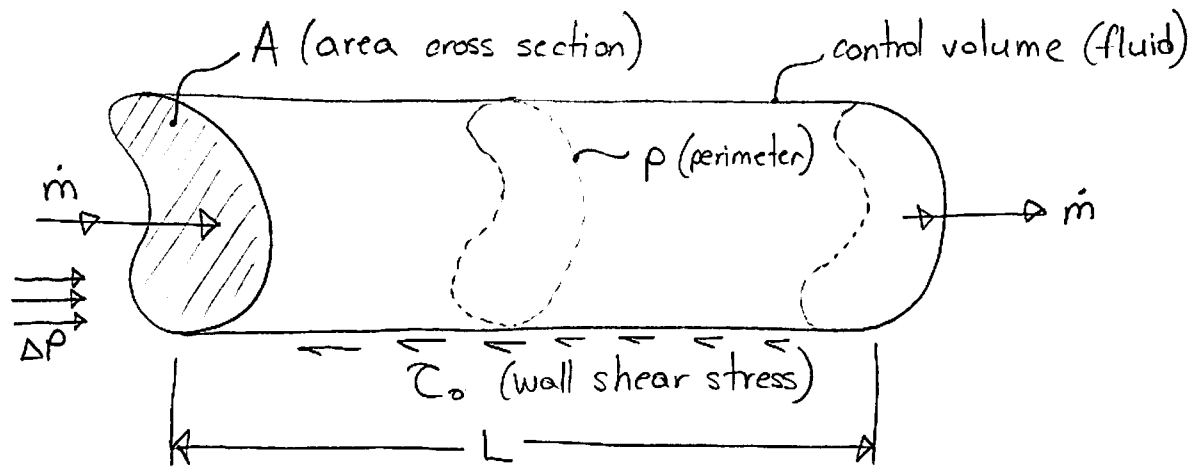


Hydraulic Diameter and Pressure Drop

The methodology of obtaining the relationship between $\dot{m}$ vs. ΔP stems from hydraulic theory of the 18th century. This is still useful today especially for turbulent flows. For our case, we can obtain an analytical result:



For a fully developed flow, we can apply a force balance on our fluid CV:

$$\sum F_x = 0 \quad (\text{since fully developed, steady})$$

$$A \Delta P = p L \tau_0 \Rightarrow \text{Note, } p \equiv \text{wetted perimeter}$$

$$\tau_0 = \frac{\Delta P}{\left(\frac{L}{A/p}\right)}$$

Here, we can define a friction factor, f , as:

$$f = \frac{\tau_0}{\frac{1}{2} \rho \bar{U}^2}$$

Note, f is similar to c_f but for fully developed flow, $f \neq f(x) = \text{constant}$
Rearranging:

$$\Delta P = f \frac{\rho L}{A} \left(\frac{1}{2} \rho \bar{U}^2 \right) \Rightarrow \text{Pressure drop across the duct.}$$

Finally, note that A/p is the linear dimension of the cross section:

$$r_h = \frac{A}{p} \equiv \text{hydraulic radius}; \quad D_h = 4r_h = \frac{4A}{p} \equiv \text{hydraulic diameter}$$

The physical meaning of D_h can be understood as the length that accounts for how close the wall (and its resistive forces - shear) are positioned relative to the stream.

We can now solve for f for our Hagen-Poiseuille flows: Let's use the round tube for simplicity:

$$\bar{u} = \frac{r_0^2}{8\mu} \left(-\frac{\partial P}{\partial x} \right) \Rightarrow \frac{8\mu \bar{u}}{r_0^2} = \frac{\Delta P}{L}$$

$$\text{We know: } f = \frac{\Delta P}{\left(\frac{4L}{D_h} \right) \frac{1}{2} \rho \bar{u}^2} \Rightarrow D_h = \frac{4A}{P} = \frac{4(\pi r_0^2)}{2\pi r_0} = 2r_0$$

$$f = \frac{\Delta P}{\left(\frac{4L}{2r_0} \right) \frac{1}{2} \rho \bar{u}^2} \Rightarrow \left(\frac{\Delta P}{L} \right) \cdot \frac{D}{2\rho \bar{u}^2} = \frac{8\mu \bar{u}}{r_0^2} \cdot \frac{D}{2\rho \bar{u}^2}$$

$$f = \frac{4\mu D}{\left(\frac{D}{2} \right)^2 \cdot \rho \bar{u}} = \frac{16\mu}{\rho \bar{u} D} = \frac{16}{Re_{Dh}}$$

$$\boxed{f = \frac{16}{Re_{Dh}}}, D_h = D \text{ (round tube)}, \boxed{Re_D = \frac{\rho \bar{u} D}{\mu}}$$

For parallel plates, we can do a similar analysis to show:

$$\boxed{f = \frac{24}{Re_{Dh}}}, D_h = 2D \text{ (D = gap thickness)}.$$

Note, these are valid for $Re_{Dh} < 2000$. Also, note the convention. We have derived the "Fanning" friction factor. There is another convention that is used for pipes:

$$\boxed{f = \frac{\Delta P}{\left(\frac{L}{D} \right) \frac{1}{2} \rho \bar{u}^2} = \frac{64}{Re_D} = \text{Darcy friction factor}}, Re_D < 2300$$

Laminar flow
Round tube.

Typically, friction factor results are tabulated as $f \cdot Re_D = \text{constant}$.

Interestingly, we can see from our previous scaling:

$$\frac{\Delta P/L}{\mu \bar{U}/D_h^2} = \frac{\frac{4f}{D_h} \left(\frac{1}{2} \rho \bar{U}^2 \right)}{\mu \bar{U}/D_h^2} = \frac{f \rho \bar{U} D_h}{\mu} = f \cdot Re_{Dh}$$

So $\boxed{\frac{\Delta P/L}{\mu \bar{U}/D_h^2} = f \cdot Re_{Dh} \sim O(1)}$ $\Rightarrow$ Makes sense. As we expected.

For many cross sections, $f \cdot Re_{Dh} \sim O(1)$ indeed.



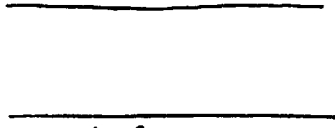
$$f \cdot Re_{Dh} = 13.3$$



$$f \cdot Re_{Dh} = 14.2$$



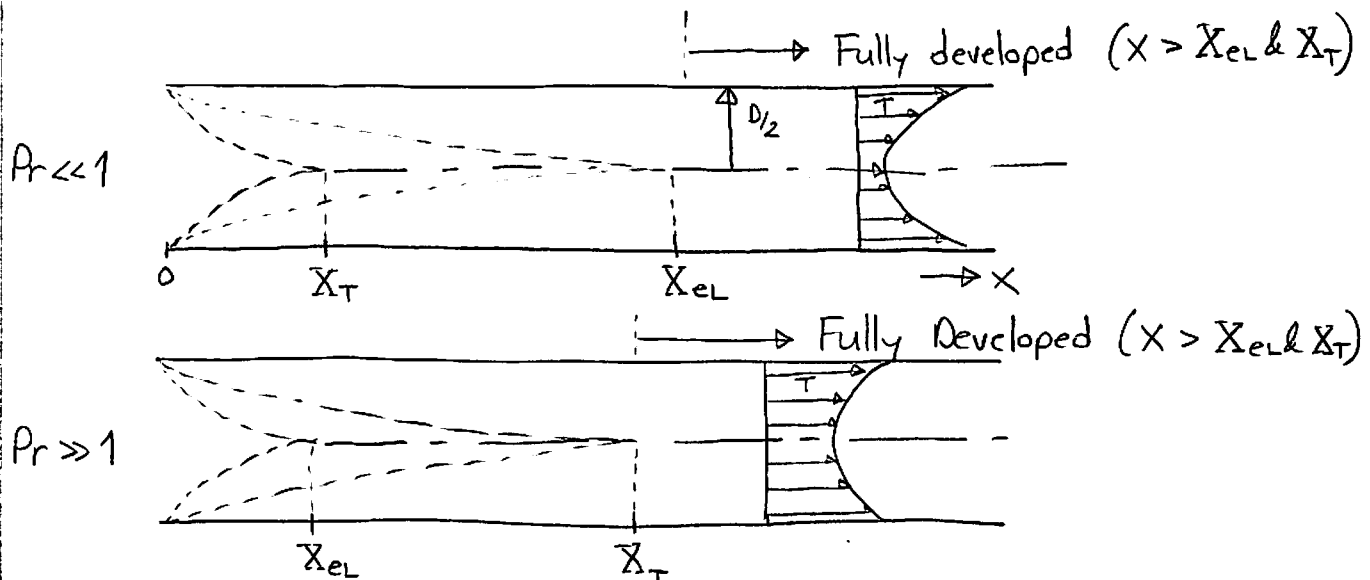
$$f \cdot Re_{Dh} = 16$$



$$f \cdot Re_{Dh} = 24$$

Heat Transfer in Fully Developed Duct Flow

To begin, we can first analyze when fully developed thermal conditions are met. We have 2 cases to consider:



For $Pr \ll 1$: $\delta_T \sim x Pr^{-1/2} Re_x^{-1/2}$ (pg. 42 of notes)

For our case, $x \sim X_T$ and $\delta_T \sim \frac{D_h}{2}$ (note, some books use $\delta_T \sim D_h$)

$$X_T Pr^{-1/2} Re_{x_T}^{-1/2} \sim \frac{D_h}{2} \Rightarrow \text{convert } Re_{x_T} \text{ to } Re_{D_h}$$

$$Re_{x_T} = \frac{\rho \bar{u} X_T}{\mu} \left(\frac{D_h}{D_h} \right) = \frac{\rho \bar{u} D_h}{\mu} \cdot \left(\frac{X_T}{D_h} \right)$$

$$X_T Pr^{-1/2} Re_{D_h}^{-1/2} \left(\frac{X_T}{D_h} \right)^{-1/2} \sim \frac{D_h}{2}$$

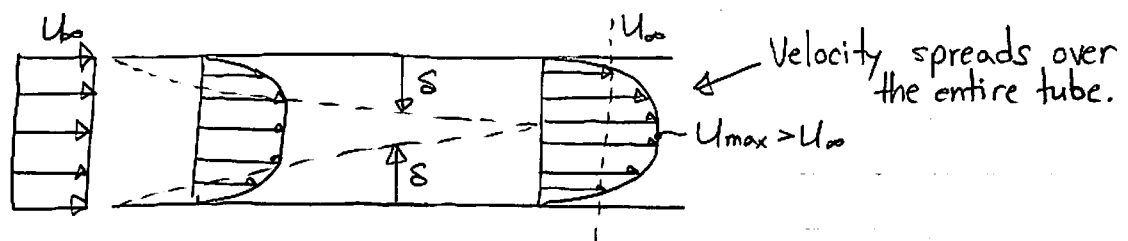
$$\boxed{\left(\frac{X_T / D_h}{Pr \cdot Re_{D_h}} \right)^{1/2} \sim \frac{1}{2}} \quad \textcircled{1} \Rightarrow \text{Thermal b.l. development}$$

Note, we could have done the same thing for the hydrodynamic b.l.:

$$\boxed{\left(\frac{X_{eL} / D_h}{Re_{D_h}} \right)^{1/2} \sim \frac{1}{2}} \quad \textcircled{2} \Rightarrow \text{Hydrodynamic b.l. development}$$

For $Pr \gg 1$: It is tempting to say that $u \sim \left(\frac{\delta_T}{\delta} \right) U_\infty$ like we did before, inside a layer of thickness δ_T . However, since our channel is confined, and the velocity scale spreads over D_h , hence $u \sim U_\infty$ and:

$$\delta_T \sim x Pr^{-1/2} Re_x^{-1/2} \Rightarrow \text{Same as } Pr \ll 1.$$



Hence:

$$\boxed{\left(\frac{X_T / D_h}{Pr \cdot Re_{D_h}} \right)^{1/2} \sim \frac{1}{2}} \Rightarrow \text{For all } Pr.$$

Dividing equations ① and ②: $\boxed{\frac{X_T}{X_{EL}} \sim Pr} \Rightarrow \text{For all } Pr.$

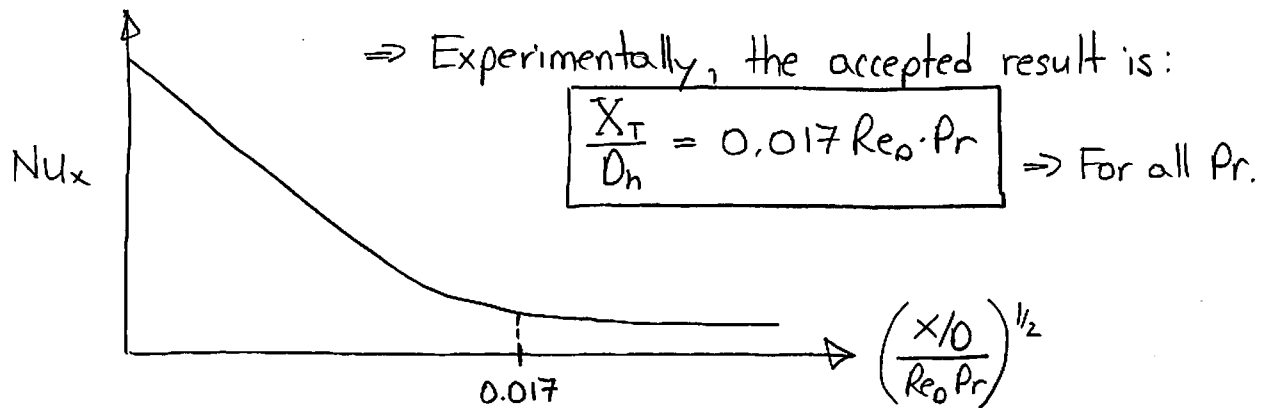
Following this up to see how heat transfer varies:

$$Nu = \frac{h D_h}{k} \sim \frac{q''}{\Delta T} \cdot \frac{D_h}{k} \sim \frac{k \frac{\Delta T}{\delta_T} \cdot D_h}{\Delta T \cdot k} \sim \frac{D_h}{\delta_T}$$

$$\delta_T \sim x Pr^{-1/2} Re_x^{-1/2} \Rightarrow \text{Convert } Re_x \text{ to } Re_{Dh}$$

$$\boxed{Nu \sim \left(\frac{x/D_h}{Re_{Dh} \cdot Pr} \right)^{-1/2}} \Rightarrow \text{Laminar} \\ \Rightarrow \text{Developing flow, for all } Pr$$

Typically our results are plotted as:



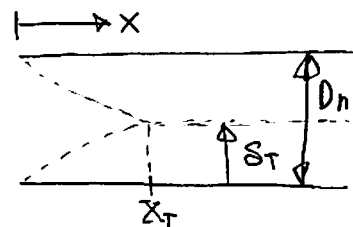
Fully Developed Region ($x > X_{EL}$ and $x > X_T$)

Before we do any analysis, let's try scaling:

$$Nu = \frac{h D_h}{k} \sim \frac{D_h}{\delta_T}$$

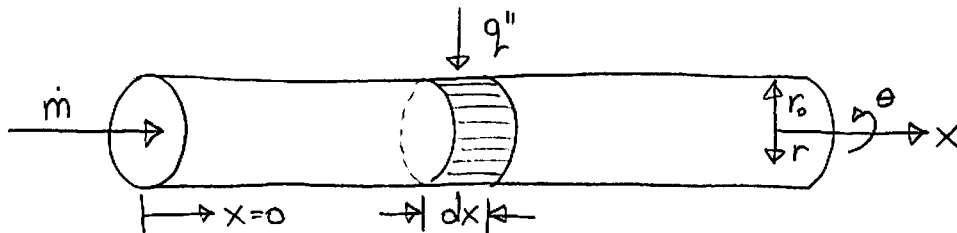
For the fully developed region, $\delta_T \sim \frac{D_h}{2}$

$$Nu \sim \frac{D_h}{D_h/2} \sim 2 \Rightarrow \boxed{Nu \sim 2}$$



We can now try to analyze the problem more exactly

Consider a pipe section of length dx undergoing heat transfer



We know that $\dot{m} = \rho \pi r_o^2 \bar{U}$

Applying an energy balance on our sliver (dx)

$$q'' \cdot 2\pi r_o dx = \dot{m}(h_{x+dx} - h_x) ; h_x = \text{fluid enthalpy at } x$$

For most fluids, we can model them as an ideal gas:

$$dh = c_p dT_m \Rightarrow \text{Back substituting}$$

$$\boxed{\frac{dT_m}{dx} = \frac{2}{r_o} \frac{q''}{\rho c_p \bar{U}}} \quad (1)$$

Note, T_m is defined as the mean temperature of the fluid as defined by thermodynamics, where the term "bulk" is used.

But we know that the fluid temperature varies as a function of x and r . However, we've defined our dh relation with respect to the thermodynamic bulk temperature.

$$q'' \cdot 2\pi r_o dx = \dot{m} dh = d \iint_A \rho u c_p T dA \Rightarrow \text{since both } u \text{ and } T \text{ are functions of } r.$$

Back substituting (1) into the equation above:

$$\frac{\rho c_p \bar{U} r_o}{2} \cdot \frac{dT_m}{dx} \cdot 2\pi r_o dx = d \iint_A \rho u c_p T dA$$

$$\int \rho c_p \pi r_o^2 \bar{U} dT_m = \int d \iiint \rho u c_p T dA$$

$$\rho c_p \bar{U} A T_m = \iint_A \rho u c_p T dA$$

Aside: for more general:

$$T_m = \frac{1}{A \bar{U}} \cdot \iint_A U T dA$$

For constant property fluids:

$$T_m = \frac{1}{\pi r_o^2 \bar{U}} \int_0^{2\pi} \int_0^{r_o} U T r dr d\theta$$

We can simplify this further since θ is usually not a factor

$$T_m = \frac{1}{\pi r_o^2 \bar{U}} \int_0^{r_o} U T 2\pi r dr \Rightarrow \text{Pipe flow, constant prop.}$$

Since the fluid temperature varies over the cross section, it is customary to define heat transfer analysis with respect to the mean temperature, T_m :

$$h = \frac{q''|_o}{T_o - T_m} = \frac{k \frac{\partial T}{\partial r}|_o}{T_o - T_m}$$

Fully Developed Temperature Profile

Let's analyze a round tube (θ -symmetric). Our energy equation becomes:

$$\frac{1}{\alpha} \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial r} \right) = \frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial x^2}$$

For the hydrodynamic and thermally fully developed region, we know that: $v=0$ and $u=u(r)$ only:

$$\underbrace{\frac{u(r)}{\alpha} \frac{\partial T}{\partial x}}_{\text{Convection}} = \underbrace{\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r}}_{\text{Conduction}} + \underbrace{\frac{\partial^2 T}{\partial x^2}}_{\text{axial}}$$

Using scaling, we can see which terms dominate and when.

We know: $u \sim \bar{U}$

$$\frac{dT_m}{dx} \sim \frac{2q''}{r_o p c_p \bar{U}} ; \quad \frac{\partial^2 T}{\partial x^2} \sim \frac{\partial^2 T_m}{\partial x^2} \sim \frac{1}{x} \left(\frac{2q''}{r_o p c_p \bar{U}} \right)$$

$$\frac{\partial^2 T}{\partial r^2} \sim \frac{\Delta T}{r_o^2} ; \quad \frac{1}{r} \frac{\partial T}{\partial r} \sim \frac{1}{r_o} \frac{\Delta T}{r_o} \sim \frac{\Delta T}{r_o^2}$$

Back substituting:

$$\frac{\bar{U}}{\alpha} \left(\frac{2q''}{r_o p c_p \bar{U}} \right) \sim \frac{\Delta T}{r_o^2} + \frac{1}{x} \left(\frac{2q''}{r_o p c_p \bar{U}} \right)$$

We know the radial conduction term must be present in order for us to solve the problem (otherwise its a trivial solution).
 Multiplying through by $r_o^2/\Delta T$

$$\frac{\bar{U}}{\alpha} \frac{r_o^2}{\Delta T} \left(\frac{2q''}{r_o p c_p \bar{U}} \right) \sim \frac{r_o^2}{\Delta T} \frac{\Delta T}{r_o^2} + \frac{1}{x} \frac{r_o^2}{\Delta T} \left(\frac{2q''}{r_o p c_p \bar{U}} \right)$$

$$\frac{hD}{k} \sim 1 + \underbrace{\frac{q''}{\Delta T}}_h \underbrace{\left(\frac{1}{r_o p c_p \bar{U}} \right)}_{\frac{\alpha}{k}} \underbrace{\left(\frac{2q''}{p c_p \bar{U}} \right)}_{\frac{\alpha}{k}} \underbrace{\left(\frac{r_o}{\Delta T} \right)}_h$$

$$\underbrace{\frac{hD}{k}}_{\text{Convection}} \sim 1 + \underbrace{\left(\frac{hD}{k} \right)^2 \left(\frac{\alpha}{\bar{U}D} \right)^2}_{\text{Conduction}}$$

Comparing our first and third scales, we see that, if

$$\frac{\bar{U}D}{\alpha} \gg 1 \Rightarrow \text{Longitudinal conduction is negligible}$$

We already know this number: $Pe_o = \frac{\bar{U}D}{\alpha} \equiv \text{Peclet \#} \equiv \frac{\text{advection}}{\text{conduction}}$

$\hookrightarrow \text{convection} = \text{conduction} + \text{advection}$

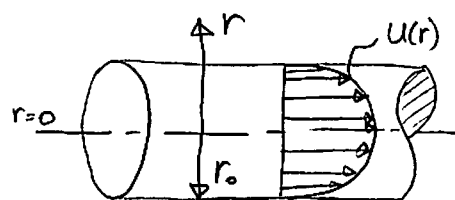
We already know from previous scaling that: $Nu = \frac{hD}{k} \sim 2$,
 here, we see that $Nu \sim 1$ (makes sense). The difference is
 in the first scaling, we used r_o instead of D for $\delta \sim r_o$.

Assuming that $Pe_0 \gg 1$, our governing equation becomes:

$$\frac{u(r)}{\alpha} \frac{\partial T}{\partial x} = \frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r}$$

sometimes written as:

$$\rho C_p u \frac{\partial T}{\partial x} = k \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right)$$



We already solved that $u = 2\bar{u} \left(1 - \frac{r^2}{r_0^2}\right) \Rightarrow$ for a round tube
Our B.C.'s are: $\left. \frac{\partial T}{\partial r} \right|_0 = 0$

$T(r=r_0) = T_0(x) \Rightarrow$ Constant heat flux case

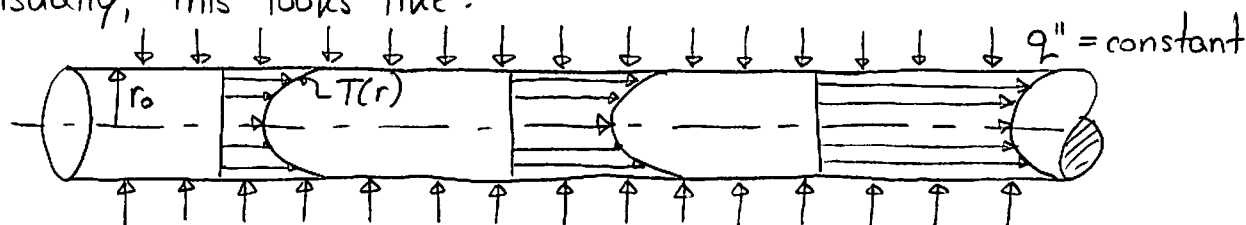
Let: $\frac{T - T_m}{T_0 - T_m} = \phi(r) \Rightarrow q''|_{r_0} = -k \left. \frac{\partial T}{\partial r} \right|_{r_0} = -k \left. \frac{\partial \phi}{\partial r} \right|_{r_0} \cdot (T_0 - T_m) = \text{constant}$

Here, we realize that the temperature profile doesn't change:

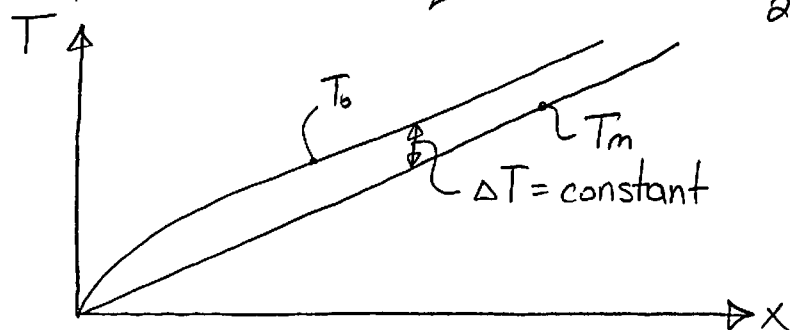
$$q''|_{r_0} = \text{constant} = \underbrace{-k \left. \frac{\partial \phi}{\partial r} \right|_{r_0}}_{\text{constant}} \underbrace{(T_0 - T_m)}_{\text{has to be constant since } \left. \frac{\partial \phi}{\partial r} \right|_{r_0} = \text{constant}} \rightarrow \frac{\partial}{\partial x} (T_0 - T_m) = \frac{\partial}{\partial x} (\text{const.})$$

$$\frac{\partial T_0}{\partial x} = \frac{\partial T_m}{\partial x}$$

Visually, this looks like:



This implies that for $q'' = \text{constant}$, $\frac{\partial T_m}{\partial x} = \frac{\partial T_0}{\partial x}$



$\Rightarrow$ if $\frac{\partial T_m}{\partial x} \neq \frac{\partial T_0}{\partial x}$, then our two lines would cross and this would violate the first law of thermodynamics.